

$$L = \int \underbrace{\sqrt{d\xi^2}}_{ds} dG \quad \epsilon \cdot dG = ds$$

$$L = \int \sqrt{\frac{\epsilon}{dG^2}} dG \Rightarrow L[\gamma] = \int \gamma \left(\frac{\epsilon}{dG^2} \right)^{\frac{1}{2}} dG = \int \gamma \epsilon^{\frac{1}{2}} = \frac{1}{2} \frac{\epsilon^{\frac{1}{2}}}{dG^2} \gamma \epsilon$$

$$\epsilon = g_{\mu\nu} dx^\mu dx^\nu \quad \frac{1}{2} \epsilon^{-1/2} \gamma \epsilon$$

$$\frac{1}{2} \epsilon^{-1/2} \gamma (g_{\mu\nu} dx^\mu dx^\nu)$$

$$\frac{1}{2} \epsilon^{-1/2} \left\{ \underbrace{\gamma g_{\mu\nu} dx^\mu dx^\nu}_{(i)} + \underbrace{\gamma g_{\mu\nu} dx^\mu dx^\nu}_{(ii)} \right\}$$

$$2 g_{\mu\nu} \gamma dx^\mu dx^\nu \quad \mu=\nu$$

$$2 g_{\mu\nu} \frac{d\gamma dx^\mu dx^\nu}{dG} \Rightarrow g_{\mu\nu} \frac{d}{dG} \left(\gamma x^\mu \frac{dx^\nu}{dG} \right) = g_{\mu\nu} \frac{d\gamma x^\mu}{dG} \frac{dx^\nu}{dG} + g_{\mu\nu} \left(\gamma x^\mu \frac{d^2 x^\nu}{dG^2} \right)$$

$$g_{\mu\nu} \frac{d\gamma x^\mu}{dG} \frac{dx^\nu}{dG} = g_{\mu\nu} \frac{d}{dG} \left(\gamma x^\mu \frac{dx^\nu}{dG} \right) - g_{\mu\nu} \left(\gamma x^\mu \frac{d^2 x^\nu}{dG^2} \right)$$

$G_{min} \rightarrow G_{max}$

$$2 g_{\mu\nu} \frac{d\gamma x^\mu}{dG} \frac{dx^\nu}{dG} = -2 g_{\mu\nu} \left(\gamma x^\mu \frac{d^2 x^\nu}{dG^2} \right) = (i)$$

$$\frac{d}{dG} (g_{\mu\nu} dx^\mu dx^\nu) = \gamma x^\mu g_{\mu\nu} \frac{d}{dG} \frac{dx^\nu}{dG} = (i)$$

$$\int \frac{1}{2} \epsilon^{-1/2} \left[\gamma x^\mu g_{\mu\nu} \frac{d}{dG} \frac{dx^\nu}{dG} - 2 g_{\mu\nu} \left(\gamma x^\mu \frac{d^2 x^\nu}{dG^2} \right) \right] dG = 0$$

$$= \int \epsilon^{-1/2} \gamma x^\mu \left[-g_{\mu\nu} \left(\frac{d^2 x^\nu}{dG^2} \right) + \frac{1}{2} g_{\mu\nu} \frac{d}{dG} \frac{dx^\nu}{dG} \right] dG = 0$$

$$- \epsilon^{-1/2} \Rightarrow \int \epsilon^{-1/2} \cdot (\epsilon^{-1/2}) \gamma x^\mu \left(g_{\mu\nu} \frac{d^2 x^\nu}{dG^2} - \frac{1}{2} g_{\mu\nu} \frac{d}{dG} \frac{dx^\nu}{dG} \right) dG = 0$$

$$\Rightarrow \int \gamma x^\mu \left(g_{\mu\nu} \left(\frac{dx^\nu}{ds} \right)^2 - \frac{1}{2} g_{\mu\nu} \frac{dx^\mu}{ds} \frac{dx^\nu}{ds} \right) ds = 0 \quad \text{Geodesik Denk}$$

$$\left(g_{\mu\nu} \left(\frac{dx^\nu}{ds} \right)^2 - \frac{1}{2} g_{\mu\nu} \frac{dx^\mu}{ds} \frac{dx^\nu}{ds} \right) = 0 = \text{Geodesik denk}$$

$$\frac{d}{ds} \left(g_{\mu\nu} \frac{dx^\nu}{ds} \right) + \frac{1}{2} \sum_{\mu\nu} \frac{dx^\mu}{ds} \frac{dx^\nu}{ds} = 0 = \text{Geodesik denk}$$

$$\sum_{\mu\nu} = \frac{1}{2} g^{\mu\nu} \left[\partial_\mu g_{\nu\sigma} + \partial_\nu g_{\sigma\mu} - \partial_\sigma g_{\mu\nu} \right]$$

$$g_{\mu\nu} = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \text{ Minkowski} \quad g_{\mu\nu} = \begin{pmatrix} 1 & 0 \\ 0 & r^2 \end{pmatrix} \quad g_{11} = g_{rr} \quad g_{22} = g_{\theta\theta}$$

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu \Rightarrow g_{11} dx^1 dx^1 + \underbrace{g_{12} dx^1 dx^2 + g_{21} dx^2 dx^1}_{0} + g_{22} dx^2 dx^2$$

$$\stackrel{\text{cyl}}{\Rightarrow} g_{\mu\nu} = g_{rr} dr dr + g_{\theta\theta} d\theta d\theta \Rightarrow dr^2 + r^2 d\theta^2 = ds^2$$

$$g_{\mu\nu} = \begin{pmatrix} 1 & 0 \\ 0 & r^2 \end{pmatrix} \Rightarrow g^{\mu\nu} = \begin{pmatrix} 1 & 0 \\ 0 & \frac{1}{r^2} \end{pmatrix} \quad g^{\mu\nu} \cdot g_{\mu\nu} = 11$$

$$\Gamma_{\mu\nu}^\lambda = \frac{1}{2} g^{\lambda\sigma} [\partial_\mu g_{\nu\sigma} + \partial_\nu g_{\sigma\mu} - \partial_\sigma g_{\mu\nu}]$$

$$\Gamma_{11}^1 \quad \Gamma_{12}^1 \quad \Gamma_{21}^1 \quad \Gamma_{22}^1 \quad \Gamma_{11}^2 \quad \Gamma_{12}^2 \quad \Gamma_{21}^2 \quad \Gamma_{22}^2$$

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 $\Gamma_{\mu\nu}^\rho = \Gamma_{\nu\mu}^\rho$

$$\Gamma_{11}^1 = \frac{1}{2} g^{11} [\underbrace{\partial_1 g_{11}}_{\frac{d}{dr} 1} + \underbrace{\partial_1 g_{11}}_{\frac{d}{dr} 1} - \underbrace{\partial_1 g_{11}}_{\frac{d}{dr} 1}] + \frac{1}{2} g^{12} [\underbrace{\partial_1 g_{12}}_0 + \underbrace{\partial_1 g_{12}}_0 - \underbrace{\partial_2 g_{11}}_0]$$

$$\Gamma_{11}^1 = 0$$

$$\Gamma_{21}^1 = \frac{1}{2} g^{11} [\underbrace{\partial_2 g_{11}}_{\frac{d}{dr} 1} + \underbrace{\partial_1 g_{12}}_0 - \underbrace{\partial_1 g_{12}}_0] + \frac{1}{2} g^{12} [\dots]$$

$$\Gamma_{21}^2 = \frac{1}{2} g^{22} [\partial_2 g_{12} + \partial_1 g_{22} - \partial_2 g_{12}]$$

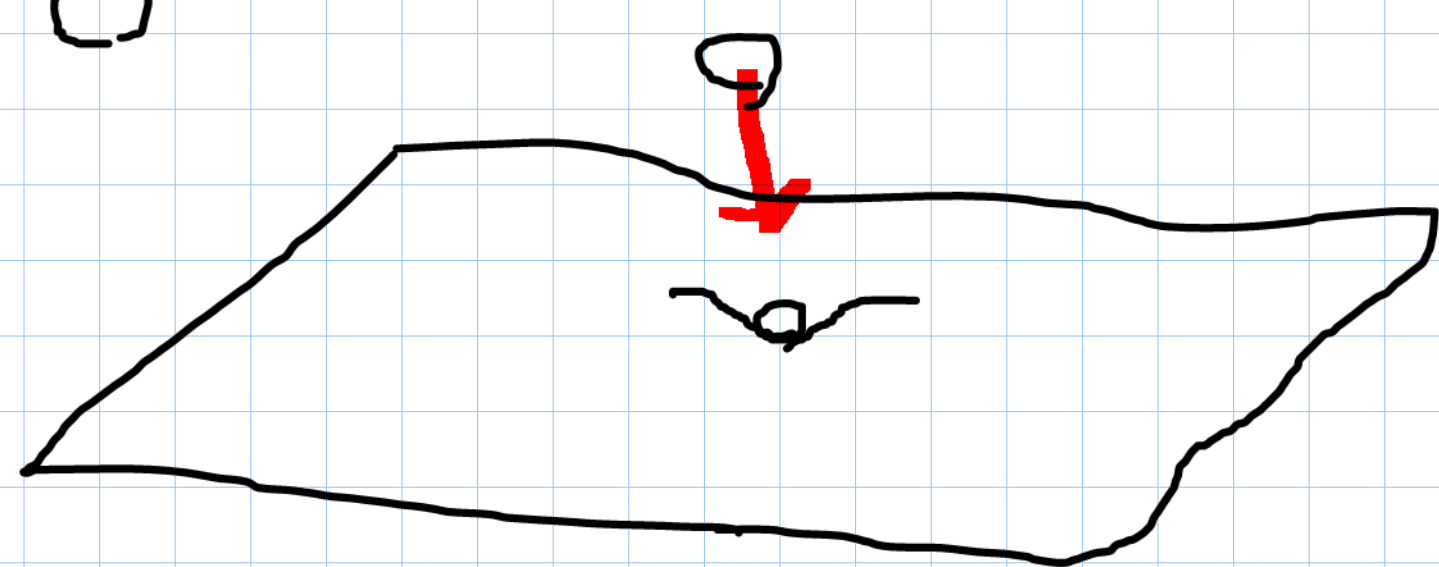
$\frac{d}{dr}(r^2) = 2r$

$$\Gamma_{21}^2 = r \quad \Gamma_{12}^2 = r$$

$$\frac{d}{ds} (g_{\mu\nu} \frac{dx^\nu}{ds}) + \frac{1}{2} \Gamma_{\mu\nu}^\lambda \frac{dx^\mu}{ds} \frac{dx^\nu}{ds} = 0$$

$$\underbrace{g_{r\theta}}_0 \frac{d}{ds} (\frac{dx^\theta}{ds}) + \frac{1}{2} (r) \frac{dx^\theta}{ds} \frac{dx^r}{ds} = \frac{r}{2} \frac{d\theta}{ds} \frac{dr}{ds}$$

8 tane Geodesik Denk Bulunur.



$$R_{jkl}^i = \frac{\partial \Gamma_{ij}^i}{\partial u_k} - \frac{\partial \Gamma_{kj}^i}{\partial u_l} + \sum_{m=1}^n (\Gamma_{km}^i \Gamma_{lj}^m - \Gamma_{lj}^m \Gamma_{km}^i)$$

$$g(R(\partial_k, \partial_l) \partial_j, \partial_m) = \sum_{i=1}^n g_{im} R_{jkl}^i$$

$$\Gamma_{21}^2 = \Gamma_{12}^2 = r \quad \Gamma_{\mu\nu}^\lambda \Rightarrow g_{im} R_{jkl}^i \Rightarrow \Gamma_{kl}^i$$

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$$\frac{\partial \Gamma_{2j}^i}{\partial u_1} - \frac{\partial \Gamma_{1j}^i}{\partial u_2} + \sum_{m=1}^2 (\Gamma_{1m}^i \Gamma_{2j}^m - \Gamma_{1j}^m \Gamma_{2m}^i)$$

$$\frac{\partial \Gamma_{2j}^i}{\partial u_1} - \frac{\partial \Gamma_{1j}^i}{\partial u_2} + \sum_{m=1}^2 \left(\Gamma_{1m}^i \Gamma_{2j}^m - \Gamma_{1j}^m \Gamma_{2m}^i \right) g_{im} R_{jkl}^i$$

$$\frac{\partial \Gamma_{21}^1}{\partial u_1} - \frac{\partial \Gamma_{11}^1}{\partial u_2} + \underbrace{\left(\Gamma_{11}^1 \Gamma_{21}^1 - \Gamma_{11}^1 \Gamma_{21}^1 \right)}_{g_{11}^{11}} + \underbrace{\left(\Gamma_{12}^1 \Gamma_{21}^2 - \Gamma_{11}^2 \Gamma_{22}^1 \right)}_{g_{12}^{11}} g_{12}^0$$

$$\frac{\partial \Gamma_{22}^1}{\partial u_1} - \frac{\partial \Gamma_{12}^1}{\partial u_2} + \left(\Gamma_{11}^1 \Gamma_{22}^1 - \Gamma_{12}^1 \Gamma_{21}^1 \right) + \left(\Gamma_{12}^2 \Gamma_{22}^2 - \Gamma_{12}^2 \Gamma_{22}^1 \right)$$

$$\boxed{R_{jkl}^i \Rightarrow g_{ip} g^{pj} R_{jkl}^i} \quad R_{jkl}^i = \frac{\partial \Gamma_{1j}^i}{\partial u_k} - \frac{\partial \Gamma_{kj}^i}{\partial u_l} + \sum_{m=1}^n \left(\Gamma_{km}^i \Gamma_{lj}^m - \Gamma_{lj}^m \Gamma_{km}^i \right)$$

$$g_{11} g^{11} R_{1kl}^1 = R_{kl}$$

$$g_{\mu\nu} = \begin{pmatrix} 1 & 0 \\ 0 & r^2 \end{pmatrix} \quad g^{\mu\nu} = \begin{pmatrix} 1 & 0 \\ 0 & \frac{1}{r^2} \end{pmatrix}$$

$$g_{22} g^{22} R_{2kl}^2$$

$$\Rightarrow \underbrace{g_{22} g^{22} R_{211}^2}_{(i)} + \underbrace{g_{22} g^{22} R_{212}^2}_{(iii)} + \underbrace{g_{22} g^{22} R_{221}^2}_{(iv)} + \underbrace{g_{22} g^{22} R_{222}^2}_{(iv)}$$

$$(i) \quad g_{22} g^{22} R_{211}^2 = \frac{\partial^2 \Gamma_{12}^2}{\partial u_1^2} - \frac{\partial^2 \Gamma_{12}^2}{\partial u_1^2} + \left(\Gamma_{11}^2 \Gamma_{12}^1 - \Gamma_{12}^1 \Gamma_{11}^2 \right) + \left(\Gamma_{12}^2 \Gamma_{12}^2 - \Gamma_{12}^2 \Gamma_{12}^2 \right)$$

$$k=1 \quad l=1 \\ i=2 \quad j=2$$

(i)+(ii)+(iii)+(iv) yapılır ve diğer eğriliklerde bulunması
lazım

$$g^{ja} g_{ia} R_{jkl}^i = \underline{R_{kl}} \Rightarrow g^{kl} R_{kl} = R$$

$$g^{kl} g^{ja} g_{ia} R_{jkl}^i = R \quad \text{skalar}$$

$$\underline{R_{ik}} - \frac{1}{2} g_{ik} R + g_{ik} \Lambda = \frac{8\pi G}{c^4} T_{ik}$$

$$R_{jkl}^i = g^{ja} g_{ia} R_{jkl}^i =$$